# TWELVE CENTURIES OF BLOOMING DATES OF THE CHERRY BLOSSOMS AT THE CITY OF KYOTO AND ITS OWN VICINITY

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*Summary* — Blooming dates of the cherry blossoms at the city of Kyoto (old capital of Japan) and its own vicinity are given in chronological order extending over twelve centuries. Examination of this documentary evidence in Japan shows a long period from 11th century until 14th century colder than the average, while the years since have been characterized by earlier blooming even though there is a definite sequence of fluctuation between late and early.

In many cases it appears that phenological data are sometimes better than temperature records themselves, as the influence of temperature is « integrated », so to speak. Study of the various data indicates that the cherry blossoms in full bloom are a good indicator of a « late » or « early » spring in Japan. The blooming date is a function of temperature, not for any one day or even for several days, but for many days. That the blooming dates give a fairly accurate record of spring character in its own vicinity is evident.

The blooming dates at the city of Kyoto (old capital city of Japan) and it own vicinity have been written by the governmental historians from the distant past. Records were not to be omitted in the earliest works on history for the Emperor or the Governor used to have a festival under the cherry blossoms in full boom almost every year. Because of its definitely decided periodicity, its uniqueness, and its gloriness, this phenomenon can be followed in descriptions going far backward through the years. Even if not a few of the old records leave the reader in doubt whether the reference is to *Prunus yedoensis*, *Prunus serrulata* or *Prunus subhirtella* and whether the reference is to efflorescent date, date a their best or some other stage, there still remains such a large number of certain records of blooming dates of the cherry-blossoms that the epochs of the climatic changes are very definitely manifest.

To show climatic history, sequences of blooming dates of the cherry blossoms at the city of Kyoto and its own vicinity may be tabulated as a chronological

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table (Table I). These data have been published in Japanese by Mr. T. TAGUCHI (1939), which constitute a unique and valuable source of climatology.

This list shows at a glance the approximate character of springs for a period of more than ten centuries. For an easier survey, the following gives the absolute extremes of the blooming dates:

Most early date ..... 1246 III 22* (1612 III 27**)
Latest date ..... 1184 V 15*

From the last compilation I make the following Table II. In it the whole period is divided into eleven century periods, and the frequencies of blooming dates are given for each decade ~III31, IV1/IV10, IV11/IV20, IV21/IV30 and V1~.

TABLE I: *Blooming dates of the cherry blossoms at the city of Kyoto and its own vicinity.*
III: March, IV: April, V: May
* the Julian calendar, ** the Gregorian calendar

| | | | |
|---|---|---|---|
| 812 III 28* | 1230 IV 5* | 1459 IV 14* | 1519 IV 11* |
| 815 IV 11* | 1246 III 22* | 1460 IV 12* | 1520 IV 4* |
| 831 IV 2* | 1247 IV 9* | 1462 IV 7* | 1521 IV 7* |
| 851 IV 14* | 1263 IV 23* | 1463 III 23* | 1522 IV 3* |
| 853 IV 10* | 1278 IV 16* | 1464 IV 4* | 1524 IV 8* |
| 864 IV 5* | 1285 IV 8* | 1465 III 30* | 1525 IV 4* |
| 866 IV 12* | 1286 IV 15* | 1466 IV 6* | 1526 IV 28* |
| | 1295 III 31* | 1468 IV 1* | 1527 IV 5* |
| 912 III 30* | | 1478 IV 2* | 1528 IV 5* |
| 917 III 31* | 1302 III 29* | 1481 IV 7* | 1529 IV 7* |
| 926 IV 3* | 1307 IV 6* | 1484 III 28* | 1530 IV 7* |
| 941 IV 14* | 1316 IV 4* | 1485 III 26* | 1531 IV 3* |
| 949 IV 12* | 1322 IV 8* | 1486 III 31* | 1532 IV 7* |
| 957 IV 17* | 1323 IV 26* | 1487 III 25* | 1585 IV 15** |
| 958 IV 14* | 1326 IV 9* | 1495 III 26* | 1594 IV 19** |
| 961 III 23* | 1331 IV 14* | 1498 III 29* | 1597 IV 24** |
| 963 III 30* | 1332 IV 16* | 1499 IV 12* | |
| 965 IV 9* | 1344 IV 3* | | 1603 IV 13** |
| 966 IV 3* | 1346 IV 9* | 1500 III 28* | 1604 IV 9** |
| 967 IV 8* | 1347 IV 11* | 1501 IV 3* | 1605 IV 15** |
| 974 IV 13* | 1366 IV 14* | 1502 V 2* | 1606 IV 16** |
| 975 IV 12* | | 1505 IV 6* | 1609 IV 18** |
| | 1422 IV 10* | 1506 IV 6* | 1612 III 27** |
| 1006 IV 4* | 1424 III 30* | 1507 IV 8* | 1613 IV 16** |
| 1016 IV 14* | 1426 III 31* | 1508 IV 10* | 1622 IV 16** |
| 1018 IV 17* | 1430 IV 9* | 1509 IV 2* | 1633 IV 8** |
| 1029 IV 20* | 1432 IV 4* | 1510 IV 10* | 1651 IV 17** |
| 1041 IV 7* | 1439 IV 15* | 1511 IV 11* | |
| | 1443 IV 8* | 1512 IV 8* | 1846 IV 7** |
| 1105 IV 17* | 1450 IV 1* | 1513 IV 10* | 1847 IV 17** |
| 1127 IV 2* | 1451 IV 4* | 1514 IV 5* | 1851 IV 12** |
| 1179 IV 6* | 1452 III 23* | 1517 IV 6* | 1853 IV 12** |
| 1184 V 15* | 1456 IV 6* | 1518 IV 15* | 1864 IV 14** |
| | 1457 IV 6* | | |
| | 1458 IV 13* | | |

Taking together these eleven century periods show the following frequency of early and late springs (percents in parentheses), after reducing into the *Gregorian calendar throughout.*

TABLE II: *Frequency of early and late spring in each century periods (percent in parentheses) in the Gregorian calendar.*

| Decade | ~ III 31 | IV 1 ~ IV 10 | IV 11 ~ IV 20 | IV 21 ~ IV 30 | V 1 ~ | Total number of obs. years | Mean |
|---|---|---|---|---|---|---|---|
| 9th century | — | 3(43) | 4(57) | — | — | 7 | IV 11 |
| 10th century | 1(7) | 5(36) | 7(50) | 1(7) | — | 14 | IV 12 |
| 11th century | — | 1(20) | 2(40) | 2(40) | — | 5 | IV 18 |
| 12th century | — | 1(25) | 1(25) | 1(25) | 1(25) | 4 | IV 24.5 |
| 13th century | 1(13) | 1(13) | 3(38) | 3(38) | — | 8 | IV 15 |
| 14th century | — | 1(8) | 7(58) | 3(25) | 1(8) | 12 | IV 18 |
| 15th century | — | 13(43) | 11(37) | 5(17) | 1(3) | 30 | IV 13 |
| 16th century | — | 1(3) | 24(77) | 4(13) | 2(6) | 31 | IV 18 |
| 17th century | 1(10) | 2(20) | 7(70) | — | — | 10 | IV 12 |
| 18th century | | | | | | 0 | |
| 19th century | — | 1(20) | 4(80) | — | — | 5 | IV 12 |
| Total | 3(2) | 29(23) | 70(56) | 19(15) | 5(4) | 126 | |
| 1917 ~ 1953 (*Prunus yedoensis*) effloresent date | 3(8) | 22(61) | 11(31) | — | — | 36 | IV 7 |
| date at their best | — | 11(31) | 21(60) | 3(9) | — | 35 | IV 14 |

Examination of this documentary evidence in Japan shows a long period between years 11th century and 14th century colder than the average. The years since have been characterized by earlier spring (i.e. earlier blooming) even though there is a definite sequence of fluctuation between late and early. Taking the figures to represent two different sets of periods, we have

$\bar{x}$ = IV 19 (for the period 11th century ~ 14th century)
$\bar{x}'$ = IV 13 (for the periods 9th and 10th centuries and 15th century ~ 19th century)
$x - \bar{x}' = +6$ days
$t = +3.0$
$n = 8$.

(R. A. FISHER: Statistical Methods for Research Workers, Oliver and Boyd, Edinburgh, Tenth Ed., 1946, § 24.1). This example shows clearly that a cold period from 11th century up to 14th century can be regarded as significant. It must be noticed that the blooming dates of *Prunus yedoensis* (normal type) are normally

later than those of other species of cherry-tree as shown in the following table given by Dr. Y. DAIGO (1947).

| Name of species | Efflorescent date at Kyoto | Number of observed years | Date at their best at Kyoto | Number of observed years |
|---|---|---|---|---|
| *Prunus serrulata* (Yama-zakura) | III 29 | 6 | IV 4 | 5 |
| *Prunus subhirtella* (Higan-zakura) | IV 5 | 9 | IV 14 | 5 |
| *Prunus yedoensis* (Somei-yoshino-zakura) | IV 8 | 26 | IV 15 | 26 |

## REFERENCES

DAIGO Y. & Y. SUZUKI, 1947: *Phenology in Japan*, Hokuryukan Book Co. (in Japanese), 217 pages. — TAGUCHI T., 1939: *Japanese Meteorological Chronology*, published by the Central Meteorological Observatory, Tokyo (in Japanese), 770 pages, with two supplementary volumes.

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